

Arbitrarily Long Finite Strings of Growing Returns

Collatz proof project

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Abstract

For every prescribed number of visits, we construct a positive natural Collatz seed whose next consecutive returns to two modulo nine all grow. Each return takes four steps with parity word 1101. This rules out a uniform fixed-return-count descent argument. The seed depends on the requested length; no infinite escaping orbit is constructed.

Use the shortcut map $T(n) = n/2$ for even n and $(3n + 1)/2$ for odd n .

Theorem 1. *For every $R \geq 0$ there is a positive $n \equiv 2 \pmod{9}$ such that its first R positive-time returns to that progression occur at times $4, 8, \dots, 4R$, and every return is larger than the preceding one. There are exactly $3R$ odd steps through time $4R$.*

Proof. If $n = 16q + 11$, exact cylinder transport gives the four iterates

$$24q + 17, \quad 36q + 26, \quad 18q + 13, \quad 27q + 20.$$

The four steps have parity word 1101, and $16T^4(n) = 27n + 23$. If also $n \equiv 2 \pmod{9}$, then $q \equiv 0 \pmod{9}$; the residues are 8, 8, 4, 2. Thus this is a first return and strictly increases n .

If 16^{R+1} divides $11n + 23$, reduction modulo sixteen forces $n \equiv 11 \pmod{16}$. The four-step identity gives

$$16(11T^4(n) + 23) = 27(11n + 23),$$

so 16^R divides $11T^4(n) + 23$. Induction therefore realizes R successive blocks whenever $16^R \mid 11n + 23$ and $n \equiv 2 \pmod{9}$.

Define $n_0 = 2$ and $n_{R+1} = 144n_R + 299$. Induction proves

$$11n_R + 23 = 45 \cdot 144^R = 16^R \cdot 45 \cdot 9^R, \quad n_R \equiv 2 \pmod{9}.$$

These positive seeds meet the required conditions. Adding the odd counts of the blocks proves the final count $3R$. $\square$

Variable branches can replenish divisibility. For a branch $F_w(n) = (A_w n + B_w)/D_w$, put $q_w(n) = (A_w - D_w)n + B_w$. Repeating the branch gives $q_w(F_w(n)) = A_w q_w(n)/D_w$, which decreases its two-adic valuation by the block length. This alone is not a termination measure when branches change. Take $w = 1101$ and $v = 101$, with respective triples $(A, B, D) = (27, 23, 16)$ and $(9, 7, 8)$. For $r \geq 1$, let $n = 16 \cdot 64^r - 5$ and $y = F_w(n)$. Lean proves the exact certificate

$$n \equiv 11 \pmod{144}, \quad y \equiv 65 \pmod{72}, \quad 11n + 23 \equiv 32 \pmod{64}, \quad y + 7 = 27 \cdot 64^r.$$

The first two congruences put the actual consecutive returns in branches w and v . Thus $v_2(q_w(n)) = 5$ but $v_2(q_v(y)) = 6r$. Even this fixed branch pair has no uniform upper bound on the valuation reset. A length-debit argument needs to account for that reset using information in the seed. This does not exclude every possible amortized potential. The new certificate is `Collatz.unbounded_reserve_reset`.

Consequence for the proof program. The coefficient through R returns is $(27/16)^R$, not a contraction. Thus no fixed number of returns guarantees descent uniformly over all positive seeds in this progression. This is compatible with the earlier sixteen-step bound for a *contracting* first return: none of these returns contract. It is also compatible with eventual contraction at a seed-dependent time. The quantifiers are $\forall R \exists n_R$, not $\exists n \forall R$; this construction does not prove divergence.

Formal verification. Run `lake build Collatz.GrowingReturns` from `lean/`. Both results use exact arithmetic, with no finite-scan premise or external axiom. No mathematical priority claim is made.